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MILITARY SPECIFICATION

SPRAY GUN, PAINT (HEAVY DUTY, SIPHON FEED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of air operated, siphon feed, heavy duty paint spray gun available in two sizes.

1.2 Classification. The spray gun covered by this specification shall be of the following sizes, as specified (see 6.2.1 and 6.5):

Size 1 - Air volume consumption not less than four cubic feet per minute (cfm) nor more than five cfm.
Size 2 - Air volume consumption not less than seven cfm nor more than eight cfm.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam.
PPP-T-60	-	Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-P-23377	-	Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-C-46168	-	Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-52192	-	Primer Coating, Epoxy.
MIL-T-81772	-	Thinner, Aircraft Coating.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-889	-	Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82	-	Standard Practice for Commercial Packaging.
ASTM E380-84	-	Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design and construction. The spray gun shall be a complete functional assembly, consisting of a gun and a cup. The spray gun shall be new and of the manufacturer's latest commercial design. The gun shall be designed for heavy duty commercial and industrial production use. The spray gun shall be for hand held operation and designed with an air pressure, fluid material atomizing and mixing cap. The spray gun shall be designed for siphon feed. The gun shall be designed and constructed to be disassembled and reassembled, for cleaning and repair, using simple hand tools without damaging the spray gun or any component. The spray gun shall be constructed of materials which are chemically compatible with the coatings, paints, solvents, and thinners encountered during life cycle use and cleaning. The internal orifice and external aligning bearing surface of the fluid needle, the fluid orifice of the air cap, and the self-aligning fluid nozzle shall be concentric with each other within 0.001 inch.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the spray guns. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Interchangeability. All parts of the spray gun shall be manufactured to definite standards and tolerances. Parts shall be interchangeable without requiring modification of the part or mating components. Parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer's part number and shall be identified in a repair parts manual (see 6.3.1). All replacement, repair, and spare parts shall provide the same or better form, fit, function, and performance as the parts provided or initially installed in the spray gun. When Industry or Government standards are applicable, all replacement parts whall conform to those standards.

3.2.3 Threads. The design, manufacture, and configuration of the threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein. No testing specified in FED-STD-H28 shall be required.

3.2.4 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when

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subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.2.5 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889. Parts in contact with the sprayable materials specified herein, except for the gun body, shall be fabricated of brass, bronze, or stainless steel.

3.3.1 Threaded aluminum interfacing. Threaded aluminum or aluminum alloy parts or components of the spray gun or cup shall not be screwed, fastened, or affixed to other male or female threaded aluminum parts or components. Oxidation and corrosion of mating (male and female) threads and fasteners can result in physical bonding of the aluminum alloy materials, which could interfere with and prevent disassembly and assembly of the spray gun.

3.3.2 Gaskets, o-rings, bushings, seals. The spray gun shall include gaskets, O-rings, bushings, or other seals to assure that there are no internal or external air, paint, or mixture leakages. All gaskets, O-rings, seals, and bushings contained in, or provided with, the spray gun shall be composed of materials which shall not undergo mechanical or chemical degradation after repeated and prolonged exposure to paints, coatings, thinners, and solvents. Non-mechanical seals such as glues, epoxies, and anaerobics shall not be used as sealants in the spray gun (see 3.4.8.4).

3.4 Performance and product characteristics. The spray gun shall be furnished complete so that, when connected to a proper power source, it can be used for the operations specified herein.

3.4.1 Sprayable materials. The spray gun shall be capable of spraying liquid materials such as enamels, lacquers, primers, shellacs, varnishes, polyurethanes, stains, paints, and two part aliphatic polyurethane coatings, conforming to MIL-P-23377, MIL-C-46168, MIL-P-52192, and MIL-T-81772, which have been thinned to a viscosity of 25 to 30 seconds using a Number 4 Ford cup at +77° Fahrenheit (F) ±5°F. The spray gun shall satisfactorily spray the materials on metal surfaces placed at any orientation to the operator.

3.4.2 Sprayable material finish. The spray gun shall be capable of applying black to charcoal color pigmented acrylic paint over a three feet by three feet square area with a coating finish equal to, or better than, standard commercial automotive passenger car finishes.

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3.4.3 Spray pattern. The spray gun shall produce a round pattern with the pattern control valve closed, and a symetrical fan spray pattern with the pattern control valve open (see Figure 1). The minimum effective dimensions of the spray pattern shall be that area of the pattern which has 100 percent coverage. The spray gun shall not produce any defective patterns, as shown in Figure 2, such as heavy bottom or top patterns, heavy left or right side patterns, or heavy center or split patterns. The spray pattern shall not change in shape when the air cap is rotated. When filled with a pigmented nitrocellulose lacquer, black to charcoal in color, having a viscosity of 25 to 30 seconds using a Number 4 Ford cup at +77° ±5° F the spray gun shall produce the fan and round patterns specified in Table I without any paint runs within the pattern area. This shall be accomplished while the gun is being held stationary and perpendicular to the spraying surface at a distance of 7.5 to 8.5 inches, inclusive, and being operated for not more than one second.

TABLE I. Spray pattern size.

Spray gun	Minimum effective dimensions of fan spray patterns (inches)		Minimum effective diameter of round spray patterns (inches)
	Major axis	Minor axis	
Size 1	6	2	2
Size 2	8	2	2

3.4.4 Air consumption. The spray gun shall produce the normal spray patterns shown in Figure 1, as specified in 3.4.3 and Table I, within the air consumption (volume), fluid flow, and air pressure limits specified in Table II.

TABLE II. Air consumption, pressure, and fluid flow.

Spray gun	Air consumption (volume, cfm)1/	Air pressure (psig)	Fluid flow (ounces per minute) (minimum)
Size 1	4-5	40-50	4.5
Size 2	7-8	50-60	8.0

1/The volume of air flow shall be corrected to 29.92 inches of mercury, standard barometric pressure, at +70° F.

3.4.5 Impact resistance. The spray gun, with the cup removed and the connections protected, shall withstand being dropped six times through a free fall of six feet to a concrete floor without damage or deformation to the spray gun, or any part or component, sufficient to degrade the performance operation of the spray gun. After each fall, the gun shall be capable of spraying as specified in 3.4.1.

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3.4.6 Operational endurance. The spray gun and all components shall withstand, without damage or deformation and perform satisfactorily after 10,000 actuations (full trigger strokes) of the spray gun trigger while being subjected to 50 to 60 psig.

3.4.7 Use simulation. The spray gun shall complete fifty spraying cycles (see 6.4.1), with a five minute interruption at mid-point in the fifty cycles, to simulate actual use conditions. A pigmented nitrocellulose lacquer shall be used.

3.4.8 Air leakage. The spray gun shall meet or exceed the requirements of paragraphs 3.4.8.1 thru 3.4.8.3.

3.4.8.1 Air leakage requirement number one. When the following pre-conditions have been fulfilled, the spray gun shall not be damaged or deformed after being subjected to air pressure of not less than 250 psig for not less than two minutes. When the air pressure supply source is shut off by a supply line valve to isolate the air pressure captured in the spray gun, the captured air pressure, 250 psig or more, in the isolated spray gun shall not drop more than five psig in not less than two minutes.

Pre-conditions of air leakage requirement number one:

(a) The spray gun trigger shall not be activated.
(b) The air and liquid material valves shall be closed to prevent the passage of air and liquid material within the spray gun body.

3.4.8.2 Air leakage requirement number two. When the following pre-conditions have been fulfilled, the spray gun shall not be damaged or deformed after being subjected to air pressure of not less than 250 psig for not less than two minutes. When the air pressure supply source is shut off by a supply line valve to isolate the air pressure captured in the spray gun, the captured air pressure, 250 psig or more, in the isolated spray gun shall not drop more than five psig in not less than two minutes.

Pre-conditions of air leakage requirement number two:

(a) The spray gun trigger shall be in a partially activated position.
(b) The air cap shall be capped, plugged, or blocked to prevent the escape of air through the air exit ports.
(c) The air valve shall be open to permit air passage within the spray gun body.
(d) The liquid material valve shall be closed to prevent liquid material passage within the spray gun body.

3.4.8.3 Air leakage requirement number three. When the following pre-conditions have been fulfilled, the spray gun shall not be damaged or deformed after being subjected to air pressure of not less than 250 psig for not less than two minutes. When the air pressure supply source is shut off by a supply line valve to isolate the air pressure captured in the spray gun, the captured air pressure, 250 psig or more, in the isolated spray gun shall not drop more than five psig in not less than two minutes.

Pre-conditions of air leakage requirement number three:

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(a) The spray trigger shall be in a fully activated position.
(b) The air cap shall be capped, plugged, or blocked to prevent the escape of air through the air exit ports.
(c) The air cap shall be capped, plugged, or blocked to prevent the escape of liquid material through the liquid material exit orifice.
(d) The air valve shall be open to permit the passage of air through the body of the spray gun.
(e) The liquid material valve shall be open to permit the passage of liquid material through the body of the spray gun.

3.4.8.4 Non-mechanical seals. As specified in 3.3.2, the spray gun shall include mechanical seals such as O-rings, gaskets, and bushings. Non-mechanical seals such as glues, epoxies, and anaerobrics shall be prohibited for use as sealants.

3.5 Detail of components. The spray gun shall consist of, but not be limited to, the components specified in 3.5.1 through 3.5.8.

3.5.1 Spray gun body. The spray gun body shall be of heavy duty, drop forged aluminum alloy. The body shall be free of fins, scale, inclusions, cold shuts, mismatching, and other defects that could affect structural integrity or designed serviceability. The gun body shall have a polished exterior surface. A hook shall be provided at the top of the body for hanging the gun when not in use.

3.5.2 Spray gun cup. A one quart capacity, attached, aluminum alloy body siphon feed cup shall be provided with each spray gun. The cup shall have a stainless steel, aluminum alloy, or zinc plated steel reinforcing ring on the bottom of the cup. The reinforcing ring shall continue not less than 0.625 inch up the side of the cup. The reinforcing ring shall be within a range of 0.042 to 0.060 inch thick. When specified (see 6.2.1), the cup interior shall be permanently coated with not less than a 2.0 mil thick polytetrafluroethylene, specifically perfluoalkoxy, coating. The adhesion of the coating to the cup interior shall be such that a strip of coating cannot be peeled from a cut edge.

3.5.3 Spray head. The spray head shall be of brass or bronze and shall be of the detachable type. The head shall be accurately machined to provide correct alignment and support of the nozzle components. The interior and exterior surfaces shall be smooth and finished with a chromium or nickel plating.

3.5.4 Air cap. The air cap shall be fabricated from brass or bronze. The caps shall have smooth exterior and interior surfaces which are chromium or nickel plated. The air cap shall be self-centering and shall be fitted to the gun head with a retainer ring or other suitable means. The air cap shall be concentric with the fluid nozzle within 0.001 inch (see 3.2).

3.5.5 Fluid nozzle. The fluid nozzle shall be of stainless steel, with a Rockwell C hardness of 48-60. The fluid nozzle shall be ground or lapped on all seating surfaces and air passages, and shall be concentric with the air cap within 0.001 inch (see 3.2). The fluid nozzle assembly shall be removeable to facilate cleaning and repair.

3.5.6 Valves. All valves shall be replaceable.

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3.5.6.1 Liquid material valve. The trigger operated liquid material valve shall be a stainless steel needle-type valve. The valve shall be ground on the seating surfaces. The needle valve packing shall be of chromed treated leather, fiber, or synthetic material. The valve shall be connected with an adjustable screw mounted at the back of the gun to permit changing the quantity of liquid material through the fluid orifice when the valve is opened. The packing shall be impervious to degradation from paints, coatings, solvents, and thinners. The liquid material valve shall be self-closing when the trigger is released.

3.5.6.2 Air valve. The trigger operated air valve shall be a cartridge-type valve or a type employing a replaceable valve seat. The valve stem packing shall be chromed treated leather, fiber, or synthetic material. The air valve shall be self-closing when the trigger is released.

3.5.6.3 Pattern control valve. A spray pattern control valve, operated from the back of the gun, shall be provided to infinitely adjust and change the spray pattern from round to fan shaped (see Figure 1).

3.5.7 Trigger. The trigger shall be designed for multiple finger control with the normal (not operator adjusted) position being closed. The trigger control shall operate freely with a positive action four to seven pound pull, and be located in front of the handle grip. The trigger shall control both the air valve and the liquid material valve, and shall operate to permit air flow before liquid material flow and liquid material flow shut-off before air flow shut-off. The trigger shall be fabricated of corrosion resistant metal or be plated to resist corrosion. The trigger shall have hardened surfaces to resist wear at points of contact with the air valve stem and the liquid material valve sleeve or stem.

3.5.8 Connections. The spray gun shall be provided with material and air connections. The connection for the liquid material shall be a 0.375 inch male standard straight pipe thread with a thirty degree female tapered seat. The connection for the air hose shall be a 0.250 inch male standard straight pipe thread with a thirty degree female tapered seat. The air connection shall be detachable and fabricated from drawn brass rod. The air hose connection shall be located in the butt of the handle grip. The liquid material connection shall be located underneath the spray gun head and be positioned to allow attachment of the spray gun cup.

3.6 Product identification. The spray gun shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number. Markings shall be on the spray gun body but not on the pressure cup.

3.7 Workmanship. Standards of workmanship shall assure that the spray gun shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specfied in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Leak tests (see 4.6.1).
(c) Performance tests (see 4.6.2).
(d) Hardness test (see 4.6.3).
(e) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105 using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or test	Level	AQL
Product examination	II	1.0
Leak tests	II	1.0
Peformance tests	II	1.0
Hardness test	II	1.0
Packaging inspection	S-4	4.0

4.5 Product examination. The spray gun shall be visually and manually examined to determine conformance with the requirements of 3.2 through 3.3.2 and 3.5 through 3.7. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of

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change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the spray guns and their components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests. Tests shall be conducted in the sequence specified in 4.6.1 through 4.6.7.

4.6.1 Leak tests. For the following leak tests, the spray gun shall be connected, in series, with an air pressure gage and a leak-free shut-off valve to a source of compressed air. The air pressure gage shall be located between the shut-off valve and the spray gun. The assembly shall be as direct and compact as possible to minimize the volume of air which can be entrapped between the shut-off valve and the spray gun. The pressure gage shall be accurate within one percent of reading over the range of 225 to 275 psig, and shall be readable to the nearest psig. The gage shall be calibrated, traceable to National Bureau of Standards calibration. The source of compressed air shall be capable of providing regulated pressures from 0 to 250 psig or more.

4.6.1.1 Valve seat leak test. With the shut-off valve open, the cup pressure regulator shall be adjusted to provide a cup pressure of 10 psig. With the spray gun trigger in the normal position, the supply pressure shall be raised to 250 psig or more. The shut-off valve shall then be closed to entrap the compressed air in the spray gun. The pressure gages shall be observed for a period of not less than two minutes. Occurance of any of the following during the test shall be cause for rejection:

(a) Visible damage to or deformation of any part of the spray gun.
(b) Air leakage from the spray gun sufficient to cause a drop in pressure of more than five psig during the two minute period.

4.6.1.2 Internal leak test. Air ports in the liquid nozzle and air cap shall be blocked by fitting the spray gun with a blank nozzle and air cap, or by other means using the standard air cap retainer or connection. With the trigger partially depressed so that the air valve is open and the liquid valve closed, the air pressure in the gun shall be raised to 250 psig. The shut-off valve shall then be closed to entrap the compressed air in the spray gun. The pressure gages shall be observed for a period of not less than two minutes. Occurrence of any of the following during the test shall be cause for rejection:

(a) Visible damage to or deformation of any part of the spray gun.
(b) Air leakage from the spray gun sufficient to cause a drop in pressure of more than five psig during the two minute period.

4.6.1.3 Valve packing leak test. The air and liquid ports in the nozzle and air cap shall be blocked by fitting the spray gun with a blank nozzle and air cap, or by other means using the standard air cap retainer or connection. With the trigger fully depressed, so that the air and liquid valves are both fully open, the air pressure in the gun shall be raised to 250 psig. The shut-off valve shall then be closed to entrap the compressed air in the spray gun. The pressure gages shall be observed for a period of not less than two minutes. Occurrence of any of the following during the test shall be cause for rejection:

(a) Visible damage to or permanent deformation of any part of the spray gun.

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(b) Air leakage from the spray gun sufficient to cause a drop in pressure of more than five psig during the two minute period.

4.6.2 Performance tests.

4.6.2.1 Spray pattern. Prior to this test, the contractor shall prepare the necessary quantity of pigmented nitrocellulose lacquer, black to charcoal in color, thinned to a viscosity of 25 to 30 seconds using a Number 4 Ford Cup at +77°±5°F. The weight of 14.5 fluid ounces ±1 fluid dram of the thinned lacquer shall then be determined to the nearest dram avoirdupois. The spray gun shall be filled with the thinned lacquer and attached, in series, with an air flow meter and an air pressure gage to a source of compressed air. The flowmeter shall be calibrated for air at 29.92 inches of mercury, standard barometric pressure, at +70°F. All readings shall be corrected for variances from those conditions. The thermometer shall be calibrated, traceable to National Bureau of Standards calibration. It shall be accurate to within .25 cfm and readable to the nearest cfm over a range of five to ten cfm. The air pressure gage shall be accurate within ±2 percent over the range of 50 to 75 psig and readable to the nearest psig. The source of compressed air shall be capable of providing regulated pressures of 0 to 100 psig over an air flow range of 0 to 10 cfm. The spray gun shall be adjusted for operation within the air pressure and air flow constraints of Table II. Liquid flow rate shall be determined by weighing the spray gun to the nearest dram avoirdupois both before and after performing a timed spray of more than one minute in duration. The spray gun shall then be operated to produce acceptable spray patterns, in accordance with 3.4.3 and Figure 1, on a vertical, flat, non-porous surface with a single burst of spray less than one second in duration for each pattern. The patterns shall include one circular pattern, two vertical fan patterns, and two horizontal fan patterns. The fan patterns shall be produced beginning with a vertical fan and continuing by rotating the air cap to 90, 180, and 270 degrees from the original position. The spray gun shall then be operated in the same manner to produce acceptable patterns on horizontal, flat, non-porous surfaces positioned both below the spray gun and above the spray gun. Following the test, the spray gun shall be cleaned in accordance with the manufacturer's instructions. Failure of the spray gun to produce acceptable patterns complying with 3.4.3 and Figure 1 within the operating constraints of 3.4.4 shall be cause for rejection.

4.6.2.2 Finish. Prior to this test, the contractor shall prepare the necessary quantity of pigmented acrylic paint, black to charcoal in color, and thinned to a viscosity of 25 to 30 seconds using a Number 4 Ford Cup at +77°±5°F. The spray gun shall be filled with the thinned acrylic and adjusted for operation as needed. The spray gun shall then be operated to completely paint three non-porous surfaces each three feet by three feet. One placed horizontally below the gun, one placed vertically, and one placed horizontally above the gun. Following this test, the gun shall be cleaned in accordance with the manufacturer's instructions. Failure of the spray gun to produce a finish as smooth or smoother over the entire surface than as specified herein (see 3.4.2) shall be cause for rejection.

4.6.3 Hardness test. The spray gun liquid nozzle shall be tested for hardness on a non-sealing surface. Failure of the nozzle to display a Rockwell C hardness of 48 to 60 shall be cause for rejection (see 3.5.5).

4.6.4 Valve endurance test. The spray gun, without paint, shall be attached to a source of regulated compressed air and adjusted for operation in accordance

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with the air flow and air pressure requirements of 3.4.4. The spray gun trigger shall be fully depressed and released for a total of 10,000 strokes (see 3.4.6). Any screws, connectors, or joints that loosen during this test shall be cause for rejection. Following this test, the spray gun shall be retested in accordance with 4.6.1 through 4.6.1.3. Failure of the spray gun to retain pressure after 10,000 actuations of the trigger or jamming of the valve shall be cause for rejection.

4.6.5 Operational endurance test. The spray gun shall be filled with pigmented nitrocellulose lacquer, black to charcoal in color, and thinned to a viscosity of 25 to 30 seconds using a Number 4 Ford Cup at +77°±5°F. The spray gun shall be adjusted for operation within the air pressure and air flow requirements of Table II. The spray gun shall then be operated through 50 spraying cycles, as specified in 3.4.7, refilling the cup with lacquer as needed. Upon completion of the fifty cycles, and without cleaning the spray gun, the spray gun shall be retested in accordance with 4.6.2 through 4.6.2.2. Failure of the spray gun to meet the performance requirements of 3.4.2 and 3.4.3 after 50 spraying cycles shall be cause for rejection.

4.6.6 Durability (impact resistance) test. The spray gun, with the cup removed, shall be emptied of paint and the air connection capped or otherwise protected. The spray gun shall then be subjected to six separate free drops through a distance of at least six feet to a bare concrete floor (see 3.4.5). Any physical damage to the spray gun, which precludes operation in accordance with 3.4.2 or 3.4.3, or any separation of components shall be cause for rejection. The spray gun shall then be retested in accordance with 4.6.1 through 4.6.2.2. Occurrence of any of the following as a result of the six drops shall be cause for rejection:

(a) Air leakage from the spray gun sufficient to permit a drop in pressure of more than five psig during any two minute period.
(b) Failure of the spray gun to meet the performance requirements of 3.4.2 and 3.4.3.

4.6.7 Maintainability test. The spray gun shall be completely disassembled and cleaned in accordance with the manufacturer's instructions and all O-rings, gaskets, washers, packings, and other seals replaced. Parts of the spray gun damaged during the durability test of 4.6.6 shall also be replaced. The spray gun shall then be reassembled and tested in accordance with 4.6.1 through 4.6.1.3 and 4.6.5. Occurrence of any of the following shall be cause for rejection:

(a) Inability to disassemble, clean, maintain, repair, and reassemble the spray gun using only common hand tools (see 3.2).
(b) Failure of the spray gun to meet the performance requirements of 3.4.
(c) Air leakage from the spray gun sufficient to permit a drop in pressure in excess of five psig during any two minute period.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Industrial, as specified (see 6.2.1).

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. All components of the spray gun and accessories shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.2 Unit package. The methods of unit protection specified herein shall conform to MIL-P-116.

5.1.1.2.1 Spray gun. Each spray gun, as defined in 3.2, shall be completely assembled and cushioned. Cushioning material shall be applied in a manner, and be of sufficient quantity, to protect and immobilize the contents of the box. Cushioning material shall conform to PPP-C-843, Type II, or PPP-C-1797 and shall be secured with tape conforming to PPP-T-60. The spray gun shall be packaged in accordance with Method IIE, using a box conforming to PPP-B-636, PPP-B-566, or PPP-B-676 as the unit container.

5.1.1.2.2 Technical data. Technical data shall be packaged in accordance with Method IC-1 and placed inside the unit container housing the spray gun.

5.1.1.3 Intermediate package. The guns shall be intermediate packaged in a fiberboard box conforming to PPP-B-636, class weather-resistant. The container shall be limited to 35 pounds gross weight and two cubic feet.

5.1.2 Industrial. Industrial preservation shall be in accordance with ASTM D3951-82. In addition, each spray gun, as defined in 3.2, shall be unit packaged completely assembled and cushioned in a fiberboard container. Cushioning shall be adequate to prevent damage from shock, vibration, stacking, or twisting. The cushioning shall be secured in place with tape. Each fiberboard container, with adequate desiccant enclosed, shall be wrapped and sealed in a water-vapor proof barrier material.

5.2 Packing. Packing shall be Level A, B, or Industrial, as specified (see 6.2.1).

5.2.1 Level A. Spray guns packaged as specified in 5.1 shall be packed in a snug-fitting box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The gross weight of each container shall be governed by the limitations of the applicable box specification. Boxes shall be uniform in size and contain like quantities as near as practicable.

5.2.2 Level B. Packing shall be the same as for Level A except the shipping containers shall conform to PPP-B-601, domestic type, style A, B, I, or J, or PPP-B-621, class 1.

5.2.3 Industrial. The preserved spray guns shall be packed in accordance with the requirements of ASTM D3951-82.

5.3 Marking.

5.3.1 Military. When military protection (Levels A or B) is specified, each unit pack, intermediate and exterior container, and unitized load shall be marked in accordance with MIL-STD-129 and any special markings required by the contract or order.

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5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of ASTM D3951-82.

6. NOTES

6.1 Intended use. This specification covers hand operated, heavy-duty spray guns intended for production use in base maintenance shops for the application of spray material by compressed air.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size spray gun required (see 1.2).
(c) First article when required (see 3.1).
(d) When a coating for the cup interior is required (see 3.5.2).
(e) Product identification if different (see 3.6).
(f) First article inspection if different (see 4.3).
(g) Quality conformance inspection if different (see 4.4).
(h) Level of protection, packing, and marking required (see 5.1, 5.2, and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name nomenclature, part number, and price (see 3.2.2). Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 Definitions.

6.4.1 Spraying cycle. A spraying cycle is defined as a single pull (engagement) of the spray gun trigger which results in the following sequential events: (1) the air valve opening to permit air passage through the spray gun; (2) the liquid material valve opening to permit the passage of air and sprayable material through the spray gun; (3) the deactivation (disengagement) of the spray gun trigger which closes the liquid material valve and the air valve to prevent the passage of liquid material and air through the spray gun; (4) with not less than two minutes elapsed from the time the trigger is engaged until the trigger is disengaged; and (5) the spray gun remains disengaged (off, non-spraying) for not less than one minute (see 3.4.7).

6.5 Supersession information. The Type 1 (pressure feed) spray gun formerly covered by this specification has been deleted from this revision.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Air Force - 99

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Navy - SH
Air Force - 82

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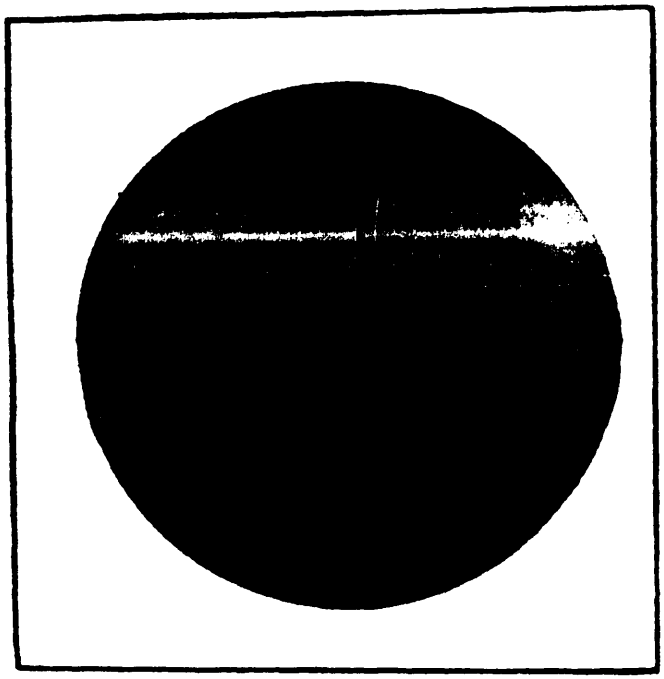
Army - ME

Preparing activity:

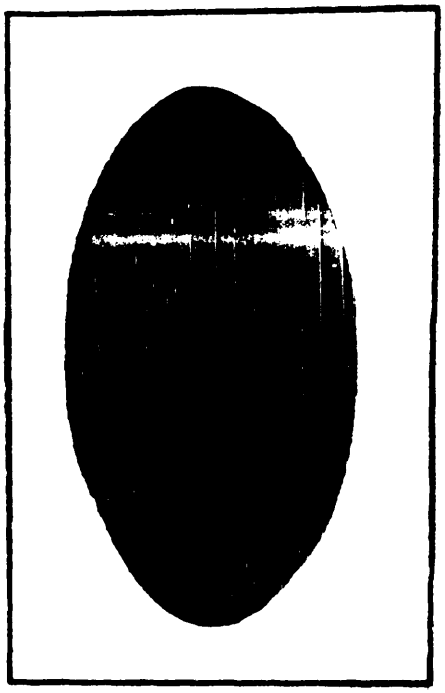
Army - AL

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Round pattern



Fan pattern

FIGURE 1. Acceptable spray patterns.

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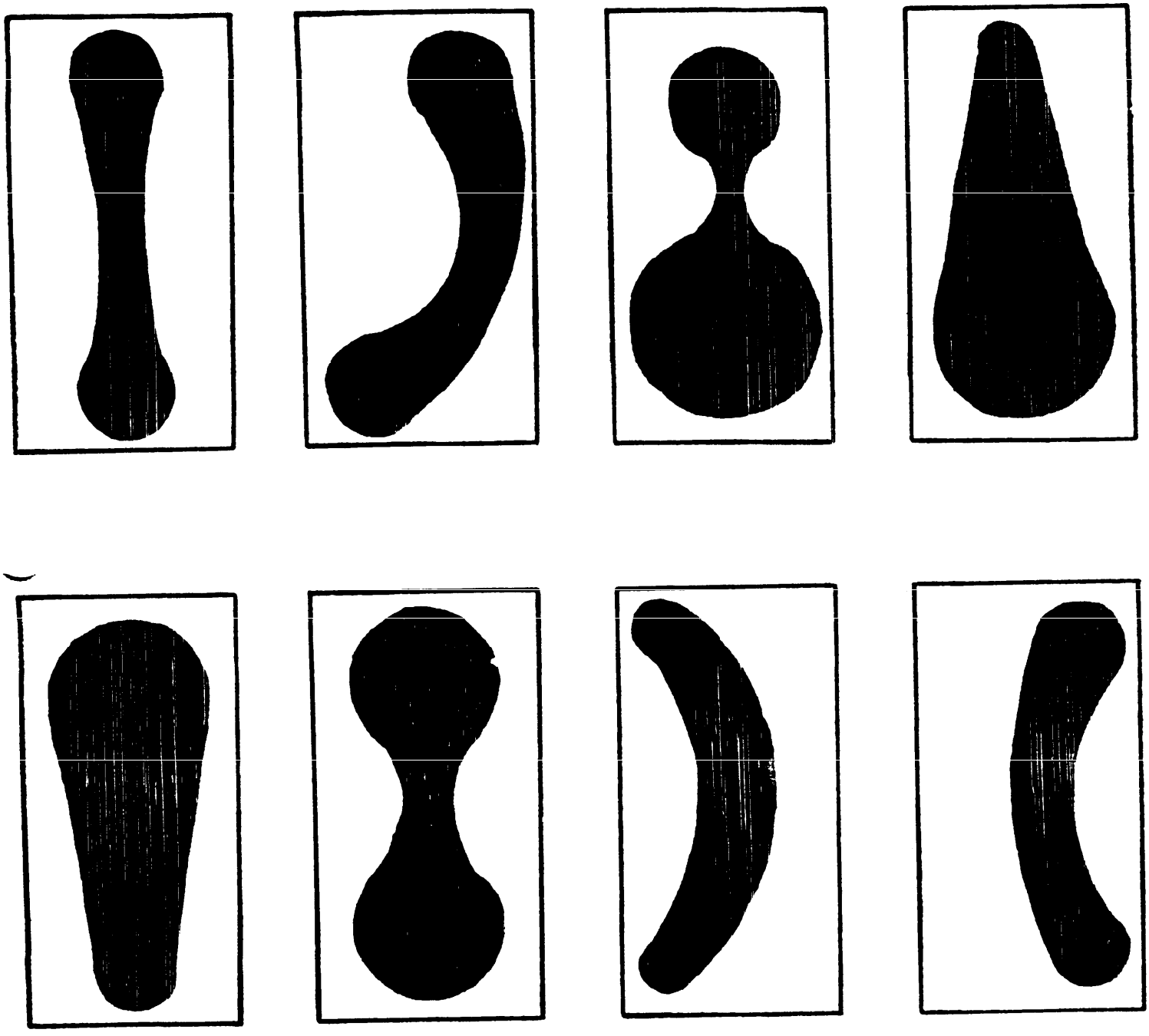


FIGURE 2. Unacceptable spray patterns.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-12877H	Spray Gun, Paint (Heavy Duty, Siphon Feed)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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